

Ten Simple Rules for Implementing Electronic Lab Notebooks (ELNs)

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S3 Appendix. Test questionnaire

This questionnaire is broadly based on the ZB MED ELN Guide (pp 45 to 49) [6].

Lab requirements

Does the ELN allow you to:

- Import and process special scientific formats?
- Export and process special scientific formats?
- Create your own templates (with drop-down menus to choose from a specified vocabulary)?
- Import templates or offer preconfigured templates? If so, which formats are supported?
- Share templates
 - With members of your own research group?
 - With external members who have a licence for the same ELN?

23 ○ With external members who have no ELN licence?

24 ● Link to files outside the ELN?

25 ● Work concurrently on the same documents as several other collaborators?

26 Does the ELN offer:

27 ● The special data entry and processing aids you need (e.g., scientific calculator, formula
28 editor, dictation devices, digital pen, animal module, antibodies module, plasmids
29 module, strains and cells module)?

30 ● Sample management and the possibility to create material databases within the ELN?

31 ● Standard interfaces (e.g., connection to a Laboratory Information Management System
32 (LIMS) or other software)?

33 ● Collaboration functions (e.g., individual users and groups, role management)?

34 Usability

35 ● Is the ELN easy to navigate?

36 ● How easily can you access a folder you have uploaded and are now looking for?

37 ● Are the options self-explanatory?

38 ● Does the ELN allow you to insert images and tables?

39 ● Does the ELN allow you to use any text formatting you want?

40 ● Does the ELN allow for freehand drawing?

41 ● Does the ELN organise data in a way that meets your needs?

42 ● Are you confident the ELN will be easy to explain to senior lab members who are used
43 to physical lab notebooks?

44 ● Are you confident the ELN will be easy to explain to new lab members?

45 Good Research Practice (GRP)

46 Does the ELN allow you to:

- 47 • Search for keywords and experiments within the ELN using free text search, advanced
48 search, database queries and special search queries (e.g., searches for chemical
49 formulas and reactions)?
- 50 • Ensure evidentiary value (e.g., audit trail, electronic signature and time stamp)?
- 51 • Easily export backups? If yes, in which formats?
- 52 • Easily export the entire ELN (e.g., for archiving purposes, or to migrate it to another
53 ELN)?
- 54 Does the ELN offer:
- 55 • Documentation and traceability in line with GSP?
- 56 • Compliance with special regulatory requirements (e.g., GxP, ISO, FERPA, HIPAA)?
- 57 • Access (at least “read only”) for a minimum of ten years?
- 58 • (In-built) archiving / backups?

59 Connection to Research Data Management (RDM) system

- 60 Does the ELN allow you to:
- 61 • Assign persistent identifiers (e.g., Digital Object Identifiers (DOIs))?
- 62 • Reference an open experiment or sample (e.g., with a Persistent Identifier (PID)), even
63 if the folder structure changes in the future?
- 64 Does the ELN offer:
- 65 • Support for assigning metadata (e.g., input masks)?
- 66 • Automatic metadata harvesting (e.g., extraction from log files, export from Excel
67 spreadsheets)?
- 68 • Integration of controlled vocabularies?
- 69 • A connection to repositories, publishing platforms, file sharing services?
- 70 • A connection to digital preservation systems?

71 IT and data security

72 Does the ELN allow you to:

- 73 • Protect sensitive data within the ELN?
- 74 • Store data locally at the institute or in the institute's own cloud?
- 75 • Run it on all operating systems (OS)?

76 Does the ELN offer:

- 77 • Application Programming Interfaces (APIs)?
- 78 • A deployment model: installation on the user's/institute's own server?
- 79 • Device-independent web-based access, also outside of your institute (e.g., via a
80 VPN)?
- 81 • An exit strategy (outgoing migration)?
- 82 • Data migration from other systems (incoming migration)?

83 Miscellaneous

84 Does the ELN offer:

- 85 • A classroom edition for teaching?
- 86 • Default language/multi-language support?

87 The following aspects are important to you:

- 88 • ...
- 89 • ...
- 90 • ...